Modeling the structural and thermal properties of loaded metal-organic frameworks. An interplay of quantum and anharmonic fluctuations

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S1 Suzuki-Chin PIMD: Partition functions, forces, virials, and stress

We present the derivations of the Suzuki-Chin partition function, the Suzuki-Chin Hamiltonian, and the expressions of the forces, the virials, and the total stress acting on the beads. All the derivations will be provided for an N-body Hamiltonian:

$$\hat{H} = \hat{T} + \hat{V} = \sum_{i=1}^{N} \frac{\hat{\mathbf{p}}_i^2}{2m_i} + V\left(\hat{\mathbf{q}}_1, \ldots, \hat{\mathbf{q}}_N, \boldsymbol{h}\right), \tag{S1.1}$$

where m_i is the mass of the i^{th} particle, V is the Born-Oppenheimer interaction potential, and $\boldsymbol{h}$ is the cell matrix of the system. The explicit dependence of the potential on the cell is henceforth dropped for ease of notation so that

$$V\left(\hat{\mathbf{q}}_1, \ldots, \hat{\mathbf{q}}_N, \boldsymbol{h}\right) \equiv V\left(\hat{\mathbf{q}}_1, \ldots, \hat{\mathbf{q}}_N\right). \tag{S1.2}$$

We also write the cell matrix as

$$\boldsymbol{h} = \mathcal{V}^{\frac{1}{3}} \boldsymbol{h}_0 \tag{S1.3}$$

in terms of the cell size (the volume $\mathcal{V}$) and the cell shape of the normalized cell matrix $\boldsymbol{h}_0$.

S1.1 The canonical partition function

The quantum canonical partition function of the system described by the Hamiltonian $\hat{H}$, at a temperature T (or $\beta = \frac{1}{k_B T}$) and a volume $\mathcal{V}$ with a fixed cell shape $\boldsymbol{h}_0$, is:

$$Z\left(N, \mathcal{V}(\boldsymbol{h}_0), T\right) = \operatorname{tr}\left(e^{-\beta \hat{H}}\right) = \operatorname{tr}\left(e^{-\beta[\hat{T}+\hat{V}]}\right). \tag{S1.4}$$

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In this form, the partition function cannot be simplified for a general interaction potential, since the exponential of the sum of the non-commuting potential and kinetic energy operators cannot be written as a product of their exponentials. One way of getting past this problem is by performing a Suzuki-Chin splitting of the Boltzmann operator at a reduced inverse temperature $\beta_P = \frac{\beta}{P}$:

$$e^{-\beta\hat{H}} = \left[e^{-\beta_P\hat{H}}\right]^P = \left[e^{-\beta_P\frac{\hat{V}_e}{3}}e^{-\beta_P\hat{T}}e^{-\beta_P\frac{4\hat{V}_o}{3}}e^{-\beta_P\hat{T}}e^{-\beta_P\frac{\hat{V}_e}{3}}\right]^{\frac{P}{2}} + \mathcal{O}\left(P^{-5}\right), \tag{S1.5}$$

with

$$\hat{V}_e = \hat{V} + \frac{\alpha}{6}\beta_P^2\,[\hat{V},[\hat{T},\hat{V}]] \qquad \text{and} \qquad \hat{V}_o = \hat{V} + \frac{(1-\alpha)}{12}\beta_P^2\,[\hat{V},[\hat{T},\hat{V}]], \tag{S1.6}$$

where $\alpha \in [0,1]$ is an arbitrary parameter which can be tuned to improve the asymptotic convergence of Equation S1.5.[1] By making use of the Suzuki-Chin splitting, an error is introduced, since the splitting is not exact at finite T, but at the same time it also allows us to write down a so-called Suzuki-Chin partition function that can be computed in the position basis:

$$\begin{aligned} Z_P^{\mathrm{sc}}(N,\mathcal{V},T) &= \mathrm{tr}\left(\left[e^{-\beta_P\frac{\hat{V}_e}{3}}e^{-\beta_P\hat{T}}e^{-\beta_P\frac{4\hat{V}_o}{3}}e^{-\beta_P\hat{T}}e^{-\beta_P\frac{\hat{V}_e}{3}}\right]^{\frac{P}{2}}\right) \\ &= \int \mathrm{d}\mathbf{q}^{(1)}\ \left\langle\mathbf{q}^{(1)}\right|\left[e^{-\beta_P\frac{\hat{V}_e}{3}}e^{-\beta_P\hat{T}}e^{-\beta_P\frac{4\hat{V}_o}{3}}e^{-\beta_P\hat{T}}e^{-\beta_P\frac{\hat{V}_e}{3}}\right]^{\frac{P}{2}}\left|\mathbf{q}^{(1)}\right\rangle, \end{aligned} \tag{S1.7}$$

where $\{\mathbf{q}^{(1)}\} \equiv \{\mathbf{q}_1^{(1)}\ldots\mathbf{q}_N^{(1)}\}$ is the position vector of a replica of the physical system. By inserting $P-1$ times the resolution of identity $\hat{I} = \int \mathrm{d}\mathbf{q}^{(j)}\ \left|\mathbf{q}^{(j)}\right\rangle\left\langle\mathbf{q}^{(j)}\right|$ between the exponentials of the kinetic and the potential energy operators, we obtain a classical partition function:

$$Z_P^{\mathrm{sc}}(N,\mathcal{V},T) = (2\pi\hbar)^{-3NP}\int \mathrm{d}\{\mathbf{p}\}\ \mathrm{d}\{\mathbf{q}\}\ \ e^{-\beta_P H_P^{\mathrm{sc}}\left(\{\mathbf{p}\},\{\mathbf{q}\},\boldsymbol{\hbar}\right)}, \tag{S1.8}$$

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where $\{\mathbf{p}\}$ and $\{\mathbf{q}\}$ are respectively shorthands for the momenta $\{\mathbf{p}^{(1)} \dots \mathbf{p}^{(P)}\}$ and the positions $\{\mathbf{q}^{(1)} \dots \mathbf{q}^{(P)}\}$ of the P replicas of the physical system and

$$H_P^{\mathrm{sc}}\left(\{\mathbf{p}\}, \{\mathbf{q}\}, \boldsymbol{h}\right) = \sum_{j=1}^{P}\left[\sum_{i=1}^{N}\frac{\left[\mathbf{p}_i^{(j)}\right]^2}{2m_i} + \sum_{i=1}^{N}\frac{1}{2}m_i\omega_P^2\left[\mathbf{q}_i^{(j)} - \mathbf{q}_i^{(j+1)}\right]^2 + V^{\mathrm{sc}}\left(\mathbf{q}^{(j)}\right)\right] \tag{S1.9}$$

is the Suzuki-Chin Hamiltonian, with $\omega_P = (\hbar\beta_P)^{-1}$. This type of Hamiltonian – that consists of P replicas of a system interacting with a spring term – is colloquially known as a ring polymer Hamiltonian and the replicas are in their turn referred to as beads. In this case, the even and odd beads feel a potential that is essentially the physical potential scaled by a factor w_j and an additional term which depends on the force:

$$\begin{aligned} V^{\mathrm{sc}}\left(\mathbf{q}^{(j)}\right) &= w_j V\left(\mathbf{q}^{(j)}\right) + w_j d_j \omega_P^{-2}\mathbf{f}^{(j)} \cdot \mathbf{u}^{(j)} \\ &= w_j V\left(\mathbf{q}^{(j)}\right) + w_j d_j \tilde{V}\left(\mathbf{q}^{(j)}\right), \end{aligned} \tag{S1.10}$$

where $\mathbf{f}^{(j)} = -\frac{\partial V\left(\mathbf{q}^{(j)}\right)}{\partial \mathbf{q}^{(j)}}$ is the force acting on the j^{th} bead, $\mathbf{u}^{(j)} = \{\frac{\mathbf{f}_1^{(j)}}{m_1} \dots \frac{\mathbf{f}_N^{(j)}}{m_N}\}$ is the acceleration of the j^{th} bead, and (w_j, d_j) are the factors that scale the effective potential felt by the beads:

$$\begin{aligned} (w_j, d_j) &= \left(\frac{2}{3}, \frac{\alpha}{6}\right) \qquad j \text{ is odd,} \\ (w_j, d_j) &= \left(\frac{4}{3}, \frac{1-\alpha}{12}\right) \quad j \text{ is even.} \end{aligned} \tag{S1.11}$$

The Suzuki-Chin partition function tends to the exact quantum canonical partition function in the limit of large P and exhibits an error of $\mathcal{O}(P^{-4})$ for finite P .

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S1.2 The isobaric-isothermal partition function

The quantum isobaric-isothermal partition function at a temperature T, cell shape $\boldsymbol{h}_0$, and pressure $\mathcal{P}$ is defined as

$$\Delta\left(N, \mathcal{P}(\boldsymbol{h}_0), T\right) \propto \int \mathrm{d}\mathcal{V}\; e^{-\beta\mathcal{P}\mathcal{V}(\boldsymbol{h}_0)}\; Z\left(N, \mathcal{V}, T\right). \tag{S1.12}$$

We perform a splitting at a high temperature (β_P) and pressure ($\mathcal{P}_P = \mathcal{P}P$):

$$\begin{aligned}\Delta\left(N, \mathcal{P}(\boldsymbol{h}_0), T\right) &\propto \int \mathrm{d}\mathcal{V}\; e^{-\beta\mathcal{P}\mathcal{V}}\; e^{-\beta\hat{H}} \\ &= \int \mathrm{d}\mathcal{V}\; e^{-\beta_P\mathcal{P}_P\mathcal{V}}\; \left[e^{-\beta_P\hat{H}}\right]^P \\ &= \int \mathrm{d}\mathcal{V}\; e^{-\beta_P\mathcal{P}_P\mathcal{V}}\; Z_P^{\mathrm{sc}}\left(N, \mathcal{V}(\boldsymbol{h}_0), T\right) + \mathcal{O}\left(P^{-4}\right) \\ &= \Delta_P^{\mathrm{sc}}\left(N, \mathcal{P}(\boldsymbol{h}_0), T\right) + \mathcal{O}\left(P^{-4}\right)\end{aligned} \tag{S1.13}$$

to obtain the Suzuki-Chin isobaric-isothermal partition function, given by:

$$\Delta_P^{\mathrm{sc}}\left(N, \mathcal{P}(\boldsymbol{h}_0), T\right) = (2\pi\hbar)^{-3NP} \int \mathrm{d}\mathcal{V} \int \mathrm{d}\{\mathbf{p}\}\; \mathrm{d}\{\mathbf{q}\}\;\; e^{-\beta_P\left[H_P^{\mathrm{sc}}\left(\{\mathbf{p}\},\{\mathbf{q}\}\right)+\mathcal{P}_P\mathcal{V}\right]}, \tag{S1.14}$$

where the configurational integral over the position of each bead is computed over the volume $\mathcal{V}$.

S1.3 The notation

Now the Suzuki-Chin Hamiltonian and the various components of the Suzuki-Chin potential are known, we proceed by explaining the notation we are going to use in the remainder of this supporting information:

- All vectors are denoted by a bold font (for instance $\mathbf{u}$) and rank-2 tensors are denoted by a sans-serif bold italic font (for instance $\boldsymbol{a}$). A superscript denotes the index of the

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replica and a subscript denotes the index of the atom.

- A vector of the form $\mathbf{a}^{(j)}$ has a length of $3N$ since it is defined for the entire system, while a vector of the form $\mathbf{a}_i^{(j)}$ has a length of 3 as it is defined for the i^{th} atom.

- To denote derivatives of the Suzuki-Chin potential, we use an additional superscript sc. The total force and virial of the j^{th} bead are for instance denoted by $\mathbf{f}^{\text{sc }(j)}$ and $\mathbf{\Xi}^{\text{sc }(j)}$.

- Since the Suzuki-Chin potential consists of a scaled physical potential and a high order term (that depends on the force), the derivatives of the potential can also be written as a weighted sum of a physical term and a high order term. The high order term will be denoted by a tilde. The total Suzuki-Chin force is hence written as: $\mathbf{f}^{\text{sc }(j)} = w_j \mathbf{f}^{(j)} + w_j d_j \tilde{\mathbf{f}}^{(j)}$. The same reasoning can be applied to the Suzuki-Chin virial, so that $\mathbf{\Xi}^{\text{sc }(j)} = w_j \mathbf{\Xi}^{(j)} + w_j d_j \tilde{\mathbf{\Xi}}^{(j)}$. Note that this is a deviation from our previous paper,[2] where the high order terms such as the potential $\tilde{V}$ did not contain the prefactor ω_P^{-2}. Thus, $\omega_P^{-2}\tilde{V}$ was in the units of energy. Please bear this in mind while making comparisons across the papers.

S1.4 The Internal Cauchy Stress

The Suzuki-Chin Helmholtz free energy at temperature T and volume $\mathcal{V}$ is given by

$$A_P^{\text{sc}} = -\beta^{-1} \log Z_P^{\text{sc}}, \tag{S1.15}$$

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so that the internal Cauchy stress (or the pressure tensor) associated with the Suzuki-Chin Hamiltonian is given by

$$\begin{aligned}
\langle \boldsymbol{P}_{\mathrm{i}}^{\mathrm{sc}} \rangle &= -\mathcal{V}^{-1} \frac{\mathrm{d}\, A_P^{\mathrm{sc}}}{\mathrm{d}\, \boldsymbol{h}} \boldsymbol{h}^T \\
&= \mathcal{V}^{-1} \beta^{-1} \frac{\mathrm{d}\, \log Z_P^{\mathrm{sc}}}{\mathrm{d}\, \boldsymbol{h}} \boldsymbol{h}^T \\
&= \mathcal{V}^{-1} \beta^{-1} Z_P^{\mathrm{sc}-1} \frac{\mathrm{d}\, Z_P^{\mathrm{sc}}}{\mathrm{d}\, \boldsymbol{h}} \boldsymbol{h}^T \\
&= \mathcal{V}^{-1} \beta^{-1} Z_P^{\mathrm{sc}-1} \left(2\pi\hbar\right)^{-3NP} \int \mathrm{d}\{\mathbf{p}\}\; \mathrm{d}\{\mathbf{q}\}\;\; \frac{\mathrm{d}}{\mathrm{d}\, \boldsymbol{h}} e^{-\beta_P H_P^{\mathrm{sc}}\left(\{\mathbf{p}\},\{\mathbf{q}\}\right)} \boldsymbol{h}^T \\
&= -\mathcal{V}^{-1} P^{-1} Z_P^{\mathrm{sc}-1} \left(2\pi\hbar\right)^{-3NP} \int \mathrm{d}\{\mathbf{p}\}\; \mathrm{d}\{\mathbf{q}\}\;\; \frac{\mathrm{d}\, H_P^{\mathrm{sc}}\left(\{\mathbf{p}\},\{\mathbf{q}\}\right)}{\mathrm{d}\, \boldsymbol{h}}\, \boldsymbol{h}^T e^{-\beta_P H_P^{\mathrm{sc}}\left(\{\mathbf{p}\},\{\mathbf{q}\}\right)}.
\end{aligned} \tag{S1.16}$$

Note that the subscript i does not refer to the atom index, but to the fact that it is the internal stress. We now perform a canonical transformation to the coordinates:

$$\mathbf{p}^{(j)} = (\boldsymbol{h}^T)^{-1} \mathbf{P}^{(j)} \quad \text{and} \quad \mathbf{q}^{(j)} = \boldsymbol{h} \mathbf{Q}^{(j)}. \tag{S1.17}$$

Since $\int \mathrm{d}\{\mathbf{P}\}\; \mathrm{d}\{\mathbf{Q}\}\; = \int \mathrm{d}\{\mathbf{p}\}\; \mathrm{d}\{\mathbf{q}\}$, we have

$$\langle \boldsymbol{P}_{\mathrm{i}}^{\mathrm{sc}} \rangle = -\mathcal{V}^{-1} P^{-1} Z_P^{\mathrm{sc}-1} \left(2\pi\hbar\right)^{-3NP} \int \mathrm{d}\{\mathbf{P}\}\; \mathrm{d}\{\mathbf{Q}\}\;\; \frac{\mathrm{d}\, H_P^{\mathrm{sc}}\left(\{\mathbf{p}\},\{\mathbf{q}\}\right)}{\mathrm{d}\, \boldsymbol{h}}\, \boldsymbol{h}^T e^{-\beta_P H_P^{\mathrm{sc}}\left(\{\mathbf{p}\},\{\mathbf{q}\}\right)}. \tag{S1.18}$$

Straightforward algebra followed by a transformation back to the Cartesian positions and momenta leads to the expression:

$$\langle \boldsymbol{P}_{\mathrm{i}}^{\mathrm{sc}} \rangle = \mathcal{V}^{-1} P^{-1} \left\langle \sum_{j=1}^{P} \left[\sum_{i=1}^{N} m_i^{-1} \mathbf{p}_i^{(j)} \otimes \mathbf{p}_i^{(j)} + \sum_{i=1}^{N} \mathbf{q}_i^{(j)} \otimes \mathbf{f}_i^{\mathrm{total}\ (j)} - \frac{\partial V^{\mathrm{sc}}\left(\mathbf{q}^{(j)}\right)}{\partial \boldsymbol{h}} \boldsymbol{h}^T \right] \right\rangle, \tag{S1.19}$$

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where $\langle \cdot \rangle$ is an average over the Suzuki-Chin canonical ensemble and

$$\mathbf{f}_i^{\text{total }(j)} = \mathbf{f}_i^{\text{spring }(j)} + \mathbf{f}_i^{\text{sc }(j)} = -m_i\omega_P^2 \left[2\mathbf{q}_i^{(j)} - \mathbf{q}_i^{(j-1)} - \mathbf{q}_i^{(j+1)}\right] - \frac{\partial V^{\text{sc}}\left(\mathbf{q}^{(j)}\right)}{\partial \mathbf{q}_i^{(j)}} \tag{S1.20}$$

is the total force, which can be written as the sum of the force coming from the spring term and the Suzuki-Chin potential. We also define the virial associated with the Suzuki-Chin potential of a given bead as

$$\mathbf{\Xi}^{\text{sc }(j)} = -\sum_{i=1}^{N} \mathbf{q}_i^{(j)} \otimes \mathbf{f}_i^{\text{sc }(j)} + \frac{\partial V^{\text{sc}}\left(\mathbf{q}^{(j)}\right)}{\partial \boldsymbol{h}} \boldsymbol{h}^T. \tag{S1.21}$$

The stress associated with the spring term can be expressed in terms of the position of the beads relative to the centroid $\bar{\mathbf{q}} = \frac{1}{P}\sum_{j=1}^{P} \mathbf{q}^{(j)}$ using the virial theorem:

$$\begin{aligned} P^{-1}\mathcal{V}^{-1}\left\langle \sum_{j=1}^{P}\sum_{i=1}^{N} \mathbf{q}_i^{(j)} \otimes \mathbf{f}_i^{\text{spring }(j)} \right\rangle &= -P^{-1}\mathcal{V}^{-1}\left\langle \sum_{j=1}^{P}\sum_{i=1}^{N} m_i\omega_P^2 \left[\mathbf{q}_i^{(j)} - \mathbf{q}_i^{(j-1)}\right] \otimes \left[\mathbf{q}_i^{(j)} - \mathbf{q}_i^{(j-1)}\right] \right\rangle \\ &= -N(P-1)\beta^{-1}\mathcal{V}^{-1}\boldsymbol{I} - \mathcal{V}^{-1}P^{-1}\left\langle \sum_{j=1}^{P}\sum_{i=1}^{N} \mathbf{f}_i^{\text{sc }(j)} \otimes \left[\mathbf{q}_i^{(j)} - \bar{\mathbf{q}}_i\right] \right\rangle. \end{aligned} \tag{S1.22}$$

The term associated with fictitious momenta of the beads is simplified to:

$$P^{-1}\mathcal{V}^{-1}\left\langle \sum_{j=1}^{P}\sum_{i=1}^{N} m_i^{-1}\mathbf{p}_i^{(j)} \otimes \mathbf{p}_i^{(j)} \right\rangle = \mathcal{V}^{-1}N\beta_P^{-1}\boldsymbol{I}. \tag{S1.23}$$

The total Cauchy stress is therefore:

$$\langle \boldsymbol{P}_{\text{i}}^{\text{sc}} \rangle = \left[\mathcal{V}^{-1}N\beta^{-1}\boldsymbol{I} - P^{-1}\mathcal{V}^{-1}\left\langle \sum_{j=1}^{P}\sum_{i=1}^{N} \mathbf{f}_i^{\text{sc }(j)} \otimes \left[\mathbf{q}_i^{(j)} - \bar{\mathbf{q}}_i\right] \right\rangle\right] - P^{-1}\mathcal{V}^{-1}\left\langle \sum_{j=1}^{P} \mathbf{\Xi}^{\text{sc }(j)} \right\rangle. \tag{S1.24}$$

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In order to compute the second term, which corresponds to the kinetic stress only, the Suzuki-Chin forces are required. The third term requires both the Suzuki-Chin forces and the derivative of the Suzuki-Chin potential with respect to the cell tensor.

S1.5 Evaluation of Suzuki-Chin forces

The instantaneous forces acting on the replicas of the Suzuki-Chin Hamiltonian

$$\begin{aligned}\mathbf{f}^{\mathrm{sc}\ (j)} &= w_j \mathbf{f}^{(j)} - w_j d_j \omega_P^{-2} \frac{\partial}{\partial \mathbf{q}^{(j)}} \left[\mathbf{f}^{(j)} \cdot \mathbf{u}^{(j)} \right] \\ &= w_j \mathbf{f}^{(j)} + w_j d_j \tilde{\mathbf{f}}^{(j)}\end{aligned} \tag{S1.25}$$

are required to perform molecular dynamics simulations which can sample the $N\mathcal{V}(\boldsymbol{h}_0)T$ and $N\mathcal{P}(\boldsymbol{h}_0)T$ ensembles. The force is expressed as a sum of two components. The first one is simply the physical force times a weight w_j that depends on the parity of the beads. The second component, which we call the high order component, depends on the Hessian and is therefore, at first sight, computationally expensive. However, using the notion that projected Hessians[3–5] can be calculated by finite differences, the high order component of the force can

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be written as:[2]

$$
\begin{aligned}
\tilde{\mathbf{f}}^{(j)} &= \omega_P^{-2}\frac{\partial}{\partial \mathbf{q}^{(j)}}\left[\sum_{i=1}^{N}\frac{\partial V\left(\mathbf{q}^{(j)}\right)}{\partial \mathbf{q}_i^{(j)}}\cdot \mathbf{u}_i^{(j)}\right] \\
&= 2\omega_P^{-2}\sum_{i=1}^{N}\left[\frac{\partial^2 V\left(\mathbf{q}^{(j)}\right)}{\partial \mathbf{q}^{(j)}\partial \mathbf{q}_i^{(j)}}\cdot \mathbf{u}_i^{(j)}\right] \\
&= 2\omega_P^{-2}\sum_{i=1}^{N}\left[\frac{\partial^2 V\left(\mathbf{q}^{(j)}\right)}{\partial \mathbf{q}_i^{(j)}\partial \mathbf{q}^{(j)}}\cdot \mathbf{u}_i^{(j)}\right] \\
&= 2\omega_P^{-2}\lim_{\epsilon\to 0}\frac{1}{\epsilon\delta}\left[\frac{\partial V\left(\mathbf{q}^{(j)}+\delta\epsilon\mathbf{u}^{(j)}\right)}{\partial \mathbf{q}^{(j)}} - \frac{\partial V\left(\mathbf{q}^{(j)}\right)}{\partial \mathbf{q}^{(j)}}\right] \\
&= -2\omega_P^{-2}\lim_{\epsilon\to 0}\frac{1}{\epsilon\delta}\left[\mathbf{f}^{(j)}\Big|_{\mathbf{q}^{(j)}+\epsilon\delta\mathbf{u}^{(j)}} - \mathbf{f}^{(j)}\right],
\end{aligned}
$$

where $\delta = \left[(3NP)^{-1}\sum_{j=1}^{P}\mathbf{u}^{(j)}\cdot\mathbf{u}^{(j)}\right]^{-\frac{1}{2}}$ is a normalization factor, so that ϵ represents the root mean square displacement applied to each atom. Thus, the Suzuki-Chin forces can be computed using an extra force evaluation of the replicas displaced in the direction of the acceleration experienced by the particles.

S1.6 Evaluation of the Suzuki-Chin virial

The viral of the effective potential felt by the replicas of the Suzuki-Chin Hamiltonian

$$
\begin{aligned}
\mathbf{\Xi}^{sc\ (j)} &= -\sum_{i=1}^{N}\mathbf{q}_i^{(j)}\otimes\mathbf{f}_i^{\mathrm{sc}\ (j)} + \frac{\partial V^{\mathrm{sc}}\left(\mathbf{q}^{(j)}\right)}{\partial \boldsymbol{h}}\boldsymbol{h}^T \\
&= w_j\mathbf{\Xi}^{(j)} + w_j d_j\tilde{\mathbf{\Xi}}^{(j)}
\end{aligned}
\tag{S1.26}
$$

is required to perform molecular dynamics simulations to sample the $N\mathscr{P}(\boldsymbol{h}_0)T$ ensemble and to estimate the instantaneous stress (or pressure). Similar to the case of the potential,

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it is expressed as a weight times the virial of the physical potential of the beads

$$\boldsymbol{\Xi}^{(j)} = -\sum_{i=1}^{N} \mathbf{q}_i^{(j)} \otimes \mathbf{f}_i^{(j)} + \frac{\partial V\left(\mathbf{q}^{(j)}\right)}{\partial \boldsymbol{h}} \boldsymbol{h}^T \tag{S1.27}$$

and that of the high order component of the Suzuki-Chin potential

$$\tilde{\boldsymbol{\Xi}}^{(j)} = -\sum_{i=1}^{N} \mathbf{q}_i^{(j)} \otimes \tilde{\mathbf{f}}_i^{(j)} + \frac{\partial \tilde{V}\left(\mathbf{q}^{(j)}\right)}{\partial \boldsymbol{h}} \boldsymbol{h}^T. \tag{S1.28}$$

The second term of S1.28 depends on the derivative of the force with respect to the cell tensor, a quantity that is generally not estimated in most codes which evaluate forces and virials. This term is also a projected second order derivative and its $\alpha\beta$ term (for instance the xx or xy term; not to be confused with $\beta = \frac{1}{k_B T}$) can be simplified as follows:

$$\begin{aligned}
\left[\frac{\partial \tilde{V}\left(\mathbf{q}^{(j)}\right)}{\partial \boldsymbol{h}} \boldsymbol{h}^T\right]_{\alpha\beta} &= \omega_P^{-2} \sum_{\gamma} \frac{\partial}{\partial h_{\alpha\gamma}} \left[\mathbf{f}^{(j)} \cdot \mathbf{u}^{(j)}\right] h_{\beta\gamma} \\
&= -\omega_P^{-2} \sum_{\gamma} \frac{\partial}{\partial h_{\alpha\gamma}} \sum_{i=1}^{N} \left[\frac{\partial V\left(\mathbf{q}^{(j)}\right)}{\partial \mathbf{q}_i^{(j)}} \cdot \mathbf{u}_i^{(j)}\right] h_{\beta\gamma} \\
&= -2\omega_P^{-2} \sum_{\gamma} \sum_{i=1}^{N} \left[\frac{\partial^2 V\left(\mathbf{q}^{(j)}\right)}{\partial h_{\alpha\gamma} \partial \mathbf{q}^{(j)}} \cdot \mathbf{u}_i^{(j)}\right] h_{\beta\gamma} \\
&= -2\omega_P^{-2} \lim_{\epsilon \to 0} \frac{1}{\epsilon\delta} \left(\sum_{\gamma} \left[\frac{\partial V\left(\mathbf{q}^{(j)} + \epsilon\delta \mathbf{u}_i^{(j)}\right)}{\partial h_{\alpha\gamma}} - \frac{\partial V\left(\mathbf{q}^{(j)}\right)}{\partial h_{\alpha\gamma}}\right] h_{\beta\gamma} \right) \\
&= -2\omega_P^{-2} \lim_{\epsilon \to 0} \frac{1}{\epsilon\delta} \left(\left[\frac{\partial V\left(\mathbf{q}^{(j)} + \epsilon\delta \mathbf{u}_i^{(j)}\right)}{\partial \boldsymbol{h}} \boldsymbol{h}^T\right]_{\alpha\beta} - \left[\frac{\partial V\left(\mathbf{q}^{(j)}\right)}{\partial \boldsymbol{h}} \boldsymbol{h}^T\right]_{\alpha\beta} \right).
\end{aligned} \tag{S1.29}$$

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The first term of S1.28, on the other hand, is simpler and can be readily written in terms of the Suzuki-Chin force computed in the previous section. However, to obtain the high order component of the virial entirely in terms of the virial of the ring polymer displaced in the direction of the acceleration of the particles, we write it as:

$$\begin{aligned}
&-\sum_{i=1}^{N}\mathbf{q}_i^{(j)}\otimes\tilde{\mathbf{f}}_i^{(j)}\\
&=2\omega_P^{-2}\lim_{\epsilon\to0}\frac{1}{\epsilon\delta}\left[\sum_{i=1}^{N}\mathbf{q}_i^{(j)}\otimes\mathbf{f}_i^{(j)}\Big|_{\mathbf{q}_i^{(j)}+\epsilon\delta\mathbf{u}_i^{(j)}}-\sum_{i=1}^{N}\mathbf{q}_i^{(j)}\otimes\mathbf{f}_i^{(j)}\right]\\
&=2\omega_P^{-2}\lim_{\epsilon\to0}\frac{1}{\epsilon\delta}\left[\sum_{i=1}^{N}\left[\mathbf{q}_i^{(j)}+\epsilon\delta\mathbf{u}_i^{(j)}\right]\otimes\mathbf{f}_i^{(j)}\Big|_{\mathbf{q}_i^{(j)}+\epsilon\delta\mathbf{u}_i^{(j)}}-\sum_{i=1}^{N}\mathbf{q}_i^{(j)}\otimes\mathbf{f}_i^{(j)}\right]-2\omega_P^{-2}\sum_{i=1}^{N}\mathbf{u}_i^{(j)}\otimes\mathbf{f}_i^{(j)}.
\end{aligned}\tag{S1.30}$$

This allows us to write the high order component of the virial as

$$\tilde{\boldsymbol{\Xi}}^{(j)}=-2\omega_P^{-2}\sum_{i=1}^{N}\mathbf{u}_i^{(j)}\otimes\mathbf{f}_i^{(j)}-2\omega_P^{-2}\lim_{\epsilon\to0}\frac{1}{\epsilon\delta}\left[\boldsymbol{\Xi}^{(j)}\Big|_{(\mathbf{q}+\epsilon\delta\mathbf{u}^{(j)})}-\boldsymbol{\Xi}^{(j)}\right].\tag{S1.31}$$

S1.7 Evaluation of the Suzuki-Chin stress tensor

Direct access to the forces and the virial allows us to compute the internal stress tensor of the Suzuki-Chin Hamiltonian:

$$\langle\boldsymbol{P}_{\mathrm{i}}^{\mathrm{sc}}\rangle=\left[\mathcal{V}^{-1}N\beta^{-1}\boldsymbol{I}-P^{-1}\mathcal{V}^{-1}\left\langle\sum_{j=1}^{P}\sum_{i=1}^{N}\mathbf{f}_i^{\mathrm{sc}\ (j)}\otimes\left[\mathbf{q}_i^{(j)}-\bar{\mathbf{q}}_i\right]\right\rangle\right]-P^{-1}\mathcal{V}^{-1}\left\langle\sum_{j=1}^{P}\boldsymbol{\Xi}^{\mathrm{sc}\ (j)}\right\rangle,\tag{S1.32}$$

where the first term is due to the fictitious masses assigned to the conjugate momenta of the ring polymer positions, the second term arises from the spring term, and the third term

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arises from the Suzuki-Chin potential. The total stress can be written as:

$$\langle \boldsymbol{P}_{\mathrm{i}}^{\mathrm{sc}} \rangle = \langle \boldsymbol{P}_{\mathrm{i}} \rangle + \langle \tilde{\boldsymbol{P}}_{\mathrm{i}} \rangle, \tag{S1.33}$$

where

$$\boldsymbol{P}_{\mathrm{i}} = \mathcal{V}^{-1} N \beta^{-1} \boldsymbol{I} - P^{-1} \mathcal{V}^{-1} \sum_{j=1}^{P} \sum_{i=1}^{N} w_j \mathbf{f}_i^{(j)} \otimes \left[\mathbf{q}_i^{(j)} - \bar{\mathbf{q}}_i \right] - P^{-1} \mathcal{V}^{-1} \sum_{j=1}^{P} w_j \boldsymbol{\Xi}^{(j)} \tag{S1.34}$$

is the stress associated with the spring term and the physical potential, with parity dependent weights applied to the potential (and its derivatives) for the different beads, and

$$\tilde{\boldsymbol{P}}_{\mathrm{i}} = -P^{-1} \mathcal{V}^{-1} \sum_{j=1}^{P} \sum_{i=1}^{N} w_j d_j \left[\tilde{\mathbf{f}}_i^{(j)} \otimes \left[\mathbf{q}_i^{(j)} - \bar{\mathbf{q}}_i \right] + \tilde{\boldsymbol{\Xi}}^{(j)} \right] \tag{S1.35}$$

is the stress associated with the high order component of the Suzuki-Chin potential. Similarly, the total internal pressure can be written as:

$$P_{\mathrm{i}}^{\mathrm{sc}} = P_{\mathrm{i}} + \tilde{P}_{\mathrm{i}} = \frac{1}{3} \mathrm{tr} \left(\boldsymbol{P}_{\mathrm{i}}^{\mathrm{sc}} \right) = \frac{1}{3} \mathrm{tr} \left(\boldsymbol{P}_{\mathrm{i}} \right) + \frac{1}{3} \mathrm{tr} \left(\tilde{\boldsymbol{P}}_{\mathrm{i}} \right). \tag{S1.36}$$

S2 Constant Pressure Suzuki-Chin PIMD: Equations of motion

In this section, we present the equations of motion to sample the $N\mathcal{P}(\boldsymbol{h}_0)T$ ensemble. The equations of motion will be written in the coordinates which allow one to write the ring polymer Hamiltonian of a free particle as that of a system of P independent harmonic

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oscillators:

$$
\begin{aligned}
H_P^0\left(\{\mathbf{p}\},\{\mathbf{q}\}\right) &= \sum_{j=1}^{P}\left[\sum_{i=1}^{N}\frac{\mathbf{p}_i^{(j)}\cdot\mathbf{p}_i^{(j)}}{2m_i}+\sum_{i=1}^{N}\frac{1}{2}m_i\omega_P^2\left[\mathbf{q}_i^{(j)}-\mathbf{q}_i^{(j+1)}\right]^2\right] \\
&= \sum_{j=1}^{P}\left[\sum_{i=1}^{N}\frac{\underset{\sim}{\mathbf{p}_i}^{(j)}\cdot\underset{\sim}{\mathbf{p}_i}^{(j)}}{2m_i}+\sum_{i=1}^{N}\frac{1}{2}m_i[\omega_P^{(j)}]^2\left[\underset{\sim}{\mathbf{q}_i}^{(j)}\right]^2\right],
\end{aligned}
\tag{S2.1}
$$

where $\omega_P^{(j)}$ are the frequencies of the normal modes as defined in Ref. 6. These new coordinates $\left(\underset{\sim}{\mathbf{p}}^{(j)},\underset{\sim}{\mathbf{q}}^{(j)}\right)$ will be referred to as normal mode coordinates. The normal mode physical and Suzuki-Chin forces are defined as $\underset{\sim}{\mathbf{f}}^{(j)}=-\frac{\partial V(\mathbf{q}^{(j)})}{\partial\underset{\sim}{\mathbf{q}}^{(j)}}$ and $\underset{\sim}{\mathbf{f}}^{\mathrm{sc}\ (\mathrm{j})}=-\frac{\partial V^{\mathrm{sc}}\ (\mathbf{q}^{(j)})}{\partial\underset{\sim}{\mathbf{q}}^{(j)}}$ respectively.

S2.1 The reduced ensemble

We proceed following the work of Martyna and co-workers[7] and target a reduced isothermal-isobaric ensemble

$$
\begin{aligned}
\Delta_P^{\mathrm{sc}}\left(N,\mathscr{P}(\boldsymbol{h}_0),T\right) &= (2\pi\hbar)^{-3NP}\int \mathrm{d}\mathcal{V}\ \int \mathrm{d}\{\mathbf{p}\}\ \mathrm{d}\{\mathbf{q}\}\ \ e^{-\beta_P\left[H_P^{\mathrm{sc}}\left(\{\mathbf{p}\},\{\mathbf{q}\}\right)+\mathscr{P}_P\mathcal{V}\right]} \\
&\approx (2\pi\hbar)^{-3NP}\int \mathrm{d}\mathcal{V}\ \int \mathrm{d}\underset{\sim}{\mathbf{p}}^{(1)}\ \mathrm{d}\underset{\sim}{\mathbf{q}}^{(1)}\ \ \int \mathrm{d}\{\underset{\sim}{\mathbf{p}}\}_{j\neq 1}\ \mathrm{d}\{\underset{\sim}{\mathbf{q}}\}_{j\neq 1}\ e^{-\beta_P\left[H_P^{\mathrm{sc}}\left(\{\mathbf{p}\},\{\mathbf{q}\}\right)+\mathscr{P}_P\mathcal{V}\right]}
\end{aligned}
\tag{S2.2}
$$

in which only the integral over the positions of the centroid (or the first normal mode) is performed over the volume $\mathcal{V}$. In other words, the cell dynamics is only coupled with that of the centroid. This ensemble was also used by Ceriotti and co-workers[8] for sampling the isothermal-isobaric ensemble with standard PIMD using the Bussi-Zykova-Parrinello barostat.[9]

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S2.2 The equations of motion

We extend the scheme of Ceriotti and co-workers[8] to the Suzuki-Chin Hamiltonian, assigning a fictitious mass μ and a momentum α to the cell. For the sake of simplicity, we assume that a PILE-L thermostat[10] is applied to the normal mode coordinates and a white noise Langevin thermostat is applied to the barostat degree of freedom. The following equations of motion lead to the conservation of the reduced $N\mathcal{P}(\boldsymbol{h}_0)T$ ensemble:

$$
\begin{aligned}
\dot{\underset{\sim}{\mathbf{p}}}_i{}^{(j)} &= \sqrt{2m_i\boldsymbol{\gamma}^{(j)}\beta_P^{-1}}\boldsymbol{\zeta}_i^{(j)} - \boldsymbol{\gamma}^{(j)}\underset{\sim}{\mathbf{p}}_i{}^{(j)} + \underset{\sim}{\mathbf{f}}_i{}^{\mathrm{sc}\ (j)} - \left[\alpha\mu^{-1}\underset{\sim}{\mathbf{p}}_i{}^{(j)}\right]\delta_{k1} - m_i[\omega_P^{(j)}]^2\underset{\sim}{\mathbf{q}}_i{}^{(j)} \\
\dot{\underset{\sim}{\mathbf{q}}}_i{}^{(j)} &= \underset{\sim}{\mathbf{p}}_i{}^{(j)}m_i^{-1} + \left[\alpha\mu^{-1}\underset{\sim}{\mathbf{q}}_i{}^{(j)}\right]\delta_{k1} \\
\dot{\mathcal{V}} &= 3\mathcal{V}\alpha\mu^{-1} \\
\dot{\alpha} &= \sqrt{2m_i\gamma^{(k)}\beta_P^{-1}}\zeta_\alpha - \gamma_\alpha\alpha + 3P\left[\left[P_{\mathrm{i}}^{\mathrm{sc}} - \mathcal{P}\right]\mathcal{V} + \beta^{-1}\right].
\end{aligned}
\tag{S2.3}
$$

Here ζ_α and $\boldsymbol{\zeta}$ correspond to a scalar and a $3NP$ dimensional random variable generated respectively from a univariate and a multivariate normal distribution. In order to have a well-defined conserved quantity, the instantaneous pressure

$$
P_{\mathrm{i}}^{\mathrm{sc}} = \frac{1}{3}P^{-1}\mathcal{V}^{-1}\mathrm{tr}\left(\sum_{i=1}^{N} m_i^{-1}\underset{\sim}{\mathbf{p}}_i{}^{(1)} \otimes \underset{\sim}{\mathbf{p}}_i{}^{(1)} - \sum_{j=1}^{P}\sum_{i=1}^{N}\mathbf{f}_i^{\mathrm{sc}\ (j)} \otimes \left[\mathbf{q}_i^{(j)} - \bar{\mathbf{q}}_i\right] - \sum_{j=1}^{P}\boldsymbol{\Xi}^{\mathrm{sc}\ (j)}\right)
\tag{S2.4}
$$

has to be used. The equations of motion have the following conserved quantity:

$$
E^{\mathrm{cons}} = H_P^{\mathrm{sc}}\left(\{\mathbf{p}\},\{\mathbf{q}\}\right) + \mathcal{P}_P\mathcal{V} + \frac{1}{2}\mu^{-1}\alpha^2 - \beta_P\log\mathcal{V} + \mathit{\Delta} H^{\mathrm{thermo}},
\tag{S2.5}
$$

where $\mathit{\Delta} H^{\mathrm{thermo}}$ is the conserved quantity associated with the heat bath.

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S2.3 The integrator

The Liouville operator that preserves the reduced $N\mathscr{P}(\boldsymbol{h}_0)T$ ensemble and that corresponds to the equations of motion in S2.3 reads:

$$i\hat{L} = i\hat{L}_\gamma + i\hat{L}_{\text{free}} + i\hat{L}_p + i\hat{L}_q + i\hat{L}_\alpha + i\hat{L}_{\mathcal{V}}. \tag{S2.6}$$

Here, the first term is the propagator of the thermostat of both the barostat and the ring polymer system. The second term corresponds to the propagation of the normal modes of the free ring polymer, while the third and the fourth term correspond to the propagation of the momentum and the position of the ring polymer. The fifth and the sixth term correspond to the propagator of the momentum and the volume of the cell. The Liouville propagators are defined as follows:

$$\begin{aligned}
i\hat{L}_\gamma &= \sum_{j=1}^{P}\sum_{i=1}^{N}\left[-\gamma_j \frac{\partial\left(\underset{\sim}{\mathbf{p}_i}^{(j)}\cdot\right)}{\partial \underset{\sim}{\mathbf{p}_i}^{(j)}} - m_i\gamma_j\beta_P^{-1}\frac{\partial^2\cdot}{\partial \underset{\sim}{\mathbf{p}_i}^{(j)2}}\right] - \gamma_j\frac{\partial\left(\alpha\cdot\right)}{\partial\alpha} - \mu\gamma_\alpha\beta_P^{-1}\frac{\partial^2\cdot}{\partial\alpha^2} \\
i\hat{L}_{\text{free}} &= \sum_{j=1}^{P}\sum_{i=1}^{N}\left[m_i^{-1}\underset{\sim}{\mathbf{p}_i}^{(j)}\frac{\partial}{\partial \underset{\sim}{\mathbf{q}_i}^{(j)}} - m_i[\omega_P^{(j)}]^2\underset{\sim}{\mathbf{q}_i}^{(j)}\frac{\partial}{\partial \underset{\sim}{\mathbf{p}_i}^{(j)}}\right] \\
i\hat{L}_p &= \sum_{j=1}^{P}\sum_{i=1}^{N}\left[\underset{\sim}{\mathbf{f}_i}^{\text{sc}\ (j)}\frac{\partial}{\partial \underset{\sim}{\mathbf{p}_i}^{(j)}}\right] \\
i\hat{L}_q &= \sum_{i=1}^{N}\left[\mu^{-1}\alpha\frac{\partial\left(\underset{\sim}{\mathbf{q}_i}^{(1)}\cdot\right)}{\partial \underset{\sim}{\mathbf{q}_i}^{(1)}}\right] \\
i\hat{L}_\alpha &= 3P\left[\mathcal{V}\left[P_{\text{i}}^{sc} - \mathscr{P}\right] + \beta^{-1}\right]\frac{\partial}{\partial\alpha} \\
i\hat{L}_{\mathcal{V}} &= 3\mu^{-1}\alpha\frac{\partial\left(\mathcal{V}\cdot\right)}{\partial\mathcal{V}}.
\end{aligned} \tag{S2.7}$$

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The integration scheme arises from a second order Trotter splitting of the Liouville operator:

$$e^{i\hat{L}\Delta t} \approx e^{i\hat{L}_\gamma \Delta t/2}\; e^{\left[i\hat{L}_p + i\hat{L}_\alpha\right]\Delta t/2}\; e^{\left[i\hat{L}_{\text{free}} + i\hat{L}_{\mathcal{V}} + i\hat{L}_q\right]\Delta t}\; e^{\left[i\hat{L}_p + i\hat{L}_\alpha\right]\Delta t/2}\; e^{i\hat{L}_\gamma \Delta t/2}. \tag{S2.8}$$

We now assume that the physical potential can be written as a sum of fast and slowly varying terms: $V(\mathbf{q}) = V^{\text{f}}(\mathbf{q}) + V^{\text{s}}(\mathbf{q})$. The corresponding forces and virials are defined as $\mathbf{f}^{\text{f/s}} = -\frac{\partial V^{\text{f/s}}(\mathbf{q})}{\partial \mathbf{q}}$ and $\mathbf{\Xi}^{\text{f/s}\,(j)} = \left[-\sum_{i=1}^{N} \mathbf{q}_i^{(j)} \otimes \mathbf{f}_i^{\text{f/s}\,(j)} + \frac{\partial V^{\text{f/s}}\left(\mathbf{q}^{(j)}\right)}{\partial \boldsymbol{h}} \boldsymbol{h}^T\right]$. We define the fast and slow components of the stress tensor as:

$$\boldsymbol{P}_{\text{i}}^{\text{f}} = \frac{1}{3} P^{-1} \mathcal{V}^{-1} \text{tr} \left(\sum_{i=1}^{N} m_i^{-1} \underset{\sim}{\mathbf{p}}_i^{(1)} \otimes \underset{\sim}{\mathbf{p}}_i^{(1)} - \sum_{j=1}^{P} \sum_{i=1}^{N} w_j \mathbf{f}_i^{\text{f}\;(j)} \otimes \left[\mathbf{q}_i^{(j)} - \bar{\mathbf{q}}_i\right] - \sum_{j=1}^{P} w_j \mathbf{\Xi}^{\text{f}\;(j)} \right) \tag{S2.9}$$

$$\boldsymbol{P}_{\text{i}}^{\text{s}} = \frac{1}{3} P^{-1} \mathcal{V}^{-1} \text{tr} \left(- \sum_{j=1}^{P} \sum_{i=1}^{N} w_j \mathbf{f}_i^{\text{s}\;(j)} \otimes \left[\mathbf{q}_i^{(j)} - \bar{\mathbf{q}}_i\right] - \sum_{j=1}^{P} w_j \mathbf{\Xi}^{\text{s}\;(j)} \right) \tag{S2.10}$$

and the scalar internal pressures are defined by simply taking the respective traces $P_i^{\text{f/s}} = \frac{1}{3}\text{tr}\left(\boldsymbol{P}_{\text{i}}^{\text{f/s}}\right)$. This allows us to split the Liouville operator for the evolution of the momenta into:

$$\begin{aligned} i\hat{L}_p &= i\hat{L}_p^{\text{f}} + i\hat{L}_p^{\text{s}} \\ i\hat{L}_\alpha &= i\hat{L}_\alpha^{\text{f}} + i\hat{L}_\alpha^{\text{s}}, \end{aligned} \tag{S2.11}$$

where

$$\begin{aligned} i\hat{L}_p^{\text{f}} &= \sum_{j=1}^{P} \sum_{i=1}^{N} \left[w_j \mathbf{f}_i^{\text{f}\;(j)} \frac{\partial}{\partial \mathbf{p}_i^{(j)}} \right] \\ i\hat{L}_\alpha^{\text{f}} &= 3P \left[\mathcal{V} \left[P_{\text{i}}^{\text{f}} - \mathcal{P} \right] + \beta^{-1} \right] \frac{\partial}{\partial \alpha} \end{aligned} \tag{S2.12}$$

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and

$$
\begin{aligned}
i\hat{L}_p^{\mathrm{s}} &= \sum_{j=1}^{P}\sum_{i=1}^{N}\left[\left(w_j \mathbf{f}_i^{\mathrm{s}\ (j)} + w_j d_j \tilde{\mathbf{f}}_i^{(j)}\right)\frac{\partial}{\partial \mathbf{p}_i^{(j)}}\right] \\
i\hat{L}_\alpha^{\mathrm{s}} &= 3P\mathcal{V}\left[P_{\mathrm{i}}^{\mathrm{s}} + \tilde{P}_{\mathrm{i}}\right]\frac{\partial}{\partial \alpha}
\end{aligned}
\tag{S2.13}
$$

It should be noted that the high order components $\tilde{\mathbf{f}}^{(j)}$ and $\tilde{\boldsymbol{\Xi}}^{(j)}$ are small because of the ω_P^{-2} pre-factor.[2] Therefore, they are included in the slow evolution operator. A multiple time step integration scheme[11] can be obtained by splitting $e^{i\hat{L}\Delta t}$ in the following way:

$$
\begin{aligned}
e^{i\hat{L}\Delta t} = \ & e^{i\hat{L}_\gamma \Delta t/2}\ e^{\left[i\hat{L}_p^{\mathrm{s}} + i\hat{L}_\alpha^{\mathrm{s}}\right]\Delta t/2}\left[e^{\left[i\hat{L}_p^{\mathrm{f}} + i\hat{L}_\alpha^{\mathrm{f}}\right]\delta t/2}\ e^{\left[i\hat{L}_{\mathrm{free}} + i\hat{L}_{\mathcal{V}} + i\hat{L}_q\right]\delta t}\ e^{\left[i\hat{L}_p^{\mathrm{f}} + i\hat{L}_\alpha^{\mathrm{f}}\right]\delta t/2}\right]^M \\
& e^{\left[i\hat{L}_p^{\mathrm{s}} + i\hat{L}_\alpha^{\mathrm{s}}\right]\Delta t/2}\ e^{i\hat{L}_\gamma \Delta t/2},
\end{aligned}
\tag{S2.14}
$$

where $\delta t = \frac{\Delta t}{M}$ is the inner time step. The thermostat is applied first for half a time step to both the ring polymer and the cell degrees of freedom. This amounts to the following update step:

$$
\begin{aligned}
\alpha &\to e^{-\gamma_\alpha \Delta t/2}\alpha + \sqrt{\mu\beta_P\left(1 - e^{-\gamma_\alpha \Delta t}\right)}\zeta_\alpha \\
\underset{\sim}{\mathbf{p}_i}^{(j)} &\to e^{-\gamma_j \Delta t/2}\underset{\sim}{\mathbf{p}_i}^{(j)} + \sqrt{m_i\beta_P\left(1 - e^{-\gamma_j \Delta t}\right)}\boldsymbol{\zeta}_i^{(j)}.
\end{aligned}
\tag{S2.15}
$$

It is followed by integrating the cell and ring polymer momenta subjected to the slow components of the forces and the virials for half a time step:

$$
\begin{aligned}
\alpha &\to \alpha + 3P\mathcal{V}\left[P_{\mathrm{i}}^{\mathrm{s}} + \tilde{P}_{\mathrm{i}}\right]\Delta t/2 \\
\mathbf{p}_i^{(j)} &\to \mathbf{p}_i{}^{(j)} + \left(w_j \mathbf{f}_i^{\mathrm{s}\ (j)} + w_j d_j \tilde{\mathbf{f}}_i^{(j)}\right)\Delta t/2.
\end{aligned}
\tag{S2.16}
$$

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We then enter the inner integration cycle where the cell and ring polymer momenta are integrated for the fast components of the forces and the virials for half the inner time step:

$$\begin{aligned} \alpha &\to \alpha + 3P\left[\mathcal{V}\left[P_{\mathrm{i}}^{\mathrm{f}} - \mathscr{P}\right] + \beta^{-1}\right]\delta t/2 \\ &+ \left[\sum_{i=1}^{N} m_i^{-1}\mathbf{F}_i^{\mathrm{f}} \cdot \underset{\sim}{\mathbf{p}_i}^{(1)}\right]\left(\delta t/2\right)^2 + \frac{1}{3}\left[\sum_{i=1}^{N} m_i^{-1}\mathbf{F}_i^{\mathrm{f}} \cdot \mathbf{F}_i^{\mathrm{f}}\right]\left(\delta t/2\right)^3 \\ \mathbf{p}_i^{(j)} &\to \mathbf{p}_i^{(j)} + w_j \mathbf{f}_i^{\mathrm{f}\ (j)}\delta t/2, \end{aligned} \tag{S2.17}$$

where $\mathbf{F}_i^{\mathrm{f}} = P^{-1}\sum_{j=1}^{P} w_j \mathbf{f}_i^{\mathrm{f}}$. This brings us to the central step, where the positions of the ring polymer and the cell volume are updated for half the inner time step:

$$\begin{aligned} \underset{\sim}{\mathbf{p}_i}^{(1)} &\to \underset{\sim}{\mathbf{p}_i}^{(1)} e^{-\mu^{-1}\alpha\delta t} \\ \underset{\sim}{\mathbf{q}_i}^{(1)} &\to \underset{\sim}{\mathbf{q}_i}^{(1)} e^{\mu^{-1}\alpha\delta t} + \underset{\sim}{\mathbf{p}_i}^{(1)} m_i^{-1}\left(\mu^{-1}\alpha\right)^{-1}\sinh\left(-\mu^{-1}\alpha\delta t\right) \\ \mathcal{V} &\to \mathcal{V} e^{3\mu^{-1}\alpha\delta t} \\ \begin{bmatrix} \underset{\sim}{\mathbf{p}_i}^{(j)} \\ \underset{\sim}{\mathbf{q}_i}^{(j)} \end{bmatrix} &\to \begin{bmatrix} \cos\omega_P^{(j)}\delta t & -m_i\omega_P^{(j)}\sin\omega_P^{(j)}\delta t \\ \left(m_i\omega_P^{(j)}\right)^{-1}\sin\omega_P^{(j)}\delta t & \cos\omega_P^{(j)}\delta t \end{bmatrix} \begin{bmatrix} \underset{\sim}{\mathbf{p}_i}^{(j)} \\ \underset{\sim}{\mathbf{q}_i}^{(j)} \end{bmatrix} \quad \forall\ \ 1 < j \leq P. \end{aligned} \tag{S2.18}$$

It is followed by updating the cell and physical momenta using the fast components for half the inner step. The inner cycle is repeated M times and is followed by the update of the cell and physical momenta using the slow components for half the time step. Finally, the thermostat step is applied again for half the time step to conclude the integration cycle of one time step. Similarly, an alternative splitting of the Liouville operator can be used to obtain a $BAOAB$-type integrator, where the thermostats are integrated in the middle:

$$\begin{aligned} e^{i\hat{L}\Delta t} \approx\ & e^{\left[i\hat{L}_p^{\mathrm{s}} + i\hat{L}_\alpha^{\mathrm{s}}\right]\Delta t/2}\left[e^{\left[i\hat{L}_p^{\mathrm{f}} + i\hat{L}_\alpha^{\mathrm{f}}\right]\delta t/2}\ e^{\left[i\hat{L}_{\mathrm{free}} + i\hat{L}_{\mathcal{V}} + i\hat{L}_q\right]\delta t}\ e^{\left[i\hat{L}_p^{\mathrm{f}} + i\hat{L}_\alpha^{\mathrm{f}}\right]\delta t/2}\right]^{M/2} \\ & e^{i\hat{L}_\gamma\Delta t}\left[e^{\left[i\hat{L}_p^{\mathrm{f}} + i\hat{L}_\alpha^{\mathrm{f}}\right]\delta t/2}\ e^{\left[i\hat{L}_{\mathrm{free}} + i\hat{L}_{\mathcal{V}} + i\hat{L}_q\right]\delta t}\ e^{\left[i\hat{L}_p^{\mathrm{f}} + i\hat{L}_\alpha^{\mathrm{f}}\right]\delta t/2}\right]^{M/2} e^{\left[i\hat{L}_p^{\mathrm{s}} + i\hat{L}_\alpha^{\mathrm{s}}\right]\Delta t/2}, \end{aligned} \tag{S2.19}$$

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which becomes for odd M:

$$\begin{aligned}e^{i\hat{L}\Delta t} \approx\ & e^{\left[i\hat{L}_p^{\mathrm{s}}+i\hat{L}_\alpha^{\mathrm{s}}\right]\Delta t/2} \left[e^{\left[i\hat{L}_p^{\mathrm{f}}+i\hat{L}_\alpha^{\mathrm{f}}\right]\delta t/2}\ e^{\left[i\hat{L}_{\mathrm{free}}+i\hat{L}_{\mathcal{V}}+i\hat{L}_q\right]\delta t}\ e^{\left[i\hat{L}_p^{\mathrm{f}}+i\hat{L}_\alpha^{\mathrm{f}}\right]\delta t/2}\right]^{(M-1)/2} \\ & e^{\left[i\hat{L}_p^{\mathrm{f}}+i\hat{L}_\alpha^{\mathrm{f}}\right]\delta t/2}\ e^{\left[i\hat{L}_{\mathrm{free}}+i\hat{L}_{\mathcal{V}}+i\hat{L}_q\right]\delta t/2}\ e^{i\hat{L}_\gamma \Delta t}\ e^{\left[i\hat{L}_{\mathrm{free}}+i\hat{L}_{\mathcal{V}}+i\hat{L}_q\right]\delta t/2}\ e^{\left[i\hat{L}_p^{\mathrm{f}}+i\hat{L}_\alpha^{\mathrm{f}}\right]\delta t/2} \\ & \left[e^{\left[i\hat{L}_p^{\mathrm{f}}+i\hat{L}_\alpha^{\mathrm{f}}\right]\delta t/2}\ e^{\left[i\hat{L}_{\mathrm{free}}+i\hat{L}_{\mathcal{V}}+i\hat{L}_q\right]\delta t}\ e^{\left[i\hat{L}_p^{\mathrm{f}}+i\hat{L}_\alpha^{\mathrm{f}}\right]\delta t/2}\right]^{(M-1)/2} e^{\left[i\hat{L}_p^{\mathrm{s}}+i\hat{L}_\alpha^{\mathrm{s}}\right]\Delta t/2}.\end{aligned} \tag{S2.20}$$

S2.4 Extension to standard (PI)MD

The MTS $N\mathscr{P}(\boldsymbol{h}_0)T$ scheme can also be applied to standard PIMD by simply dropping all the high order terms (*i.e.* setting $d_j = 0$) and setting the physical weights w_j to 1. Similarly, by setting the number of replicas to 1, the scheme can be used to sample the classical isothermal-isobaric ensemble with multiple time stepping and $BAOAB$-type integrators.

S3 Heat capacity estimators

In order to define the heat capacity, we first define the (various) estimators of the energy and the enthalpy. The primitive estimator of the energy in terms of the partition function is given by

$$\begin{aligned}\left\langle \mathcal{E}^{\mathrm{PR}} \right\rangle &= -\frac{1}{Z_P^{\mathrm{sc}}\left(N,\mathcal{V},T\right)} \frac{\partial Z_P^{\mathrm{sc}}\left(N,\mathcal{V},T\right)}{\partial \beta} \\ &= \left\langle \frac{3NP}{2\beta} + \frac{1}{P}\sum_{j=1}^{P}\left[-\sum_{i=1}^{N}\frac{1}{2}m_i\omega_P^2\left[\mathbf{q}_i^{(j)}-\mathbf{q}_i^{(j+1)}\right]^2 + w_j V(\mathbf{q}^{(j)}) + 3w_j d_j \tilde{V}(\mathbf{q}^{(j)})\right]\right\rangle .\end{aligned} \tag{S3.1}$$

This estimator is highly inefficient since the spring term has a variance that grows with P. A more efficient centroid-virial thermodynamic estimator is obtained by performing an

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integration by parts:

$$\left\langle \mathcal{E}^{\mathrm{TD}} \right\rangle = \left\langle \frac{3N}{2\beta} + \frac{1}{P} \sum_{j=1}^{P} \left[-\frac{1}{2} \mathbf{f}^{\mathrm{sc}\ (j)} \cdot \left[\mathbf{q}^{(j)} - \bar{\mathbf{q}} \right] + w_j V(\mathbf{q}^{(j)}) + 3 w_j d_j \tilde{V}(\mathbf{q}^{(j)}) \right] \right\rangle . \quad \text{(S3.2)}$$

Within the Suzuki-Chin scheme, there exists a simpler way of estimating the average energy. Instead of deriving the partition function with respect to β, one can simply compute the expectation value of the energy operator. This leads to a so-called centroid-virial operator estimator, that reads:

$$\left\langle \mathcal{E}^{\mathrm{OP}} \right\rangle = \left\langle \frac{3N}{2\beta} + \frac{2}{P} \sum_{j=1}^{P/2} \left[-\frac{1}{2} \mathbf{f}^{(2j-1)} \cdot \left[\mathbf{q}^{(2j-1)} - \bar{\mathbf{q}} \right] + V(\mathbf{q}^{(2j-1)}) \right] \right\rangle , \quad \text{(S3.3)}$$

which only depends on the physical forces of the odd beads. The corresponding estimators of the enthalpy are defined as

$$\left\langle \mathcal{H}^{\square} \right\rangle = \left\langle \mathcal{E}^{\square} + \mathcal{P}\mathcal{V} \right\rangle . \quad \text{(S3.4)}$$

Various estimators of the isochoric heat capacity can be obtained by deriving the corresponding estimators of the energy:

$$\frac{C_V^{\square}}{k_B \beta^2} = \left\langle \mathcal{E}^{\square} \cdot \mathcal{E}^{\mathrm{PR}} \right\rangle - \left\langle \mathcal{E}^{\square} \right\rangle \cdot \left\langle \mathcal{E}^{\mathrm{PR}} \right\rangle - \left\langle \frac{\partial \mathcal{E}^{\square}}{\partial \beta} \right\rangle . \quad \text{(S3.5)}$$

These estimators are inefficient because of the poor statistical properties of the primitive estimators of the energy. An improvement can be made by performing another integration by parts,[12] which gives:

$$\frac{C_V^{\square}}{k_B \beta^2} = \left\langle \mathcal{E}^{\square} \cdot \mathcal{E}^{\mathrm{TD}} \right\rangle - \left\langle \mathcal{E}^{\square} \right\rangle \cdot \left\langle \mathcal{E}^{\mathrm{TD}} \right\rangle + \left\langle \mathcal{E}^{\square\prime} \right\rangle , \quad \text{(S3.6)}$$

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where

$$\mathcal{E}^{\square\prime} = -\left\langle \frac{\partial \mathcal{E}^{\square}}{\partial \beta} \right\rangle - \frac{1}{2\beta} \left\langle \sum_{j=1}^{P} (\mathbf{q}^{(j)} - \bar{\mathbf{q}}) \frac{\partial \mathcal{E}^{\square}}{\partial \mathbf{q}^{(j)}} \right\rangle . \tag{S3.7}$$

The estimators created through this process are called double virial estimators. These estimators have a variance that does not explicitly depend on the number of beads P. The $\mathcal{E}^{\square\prime}$ term is however computationally expensive, as it depends on the higher order derivatives of the potential. Similarly, the estimators of the isobaric heat capacity are given by

$$\frac{C_P^{\square}}{k_B \beta^2} = \left\langle \mathcal{H}^{\square} \cdot \mathcal{H}^{\mathrm{TD}} \right\rangle - \left\langle \mathcal{H}^{\square} \right\rangle \cdot \left\langle \mathcal{H}^{\mathrm{TD}} \right\rangle + \left\langle \mathcal{E}^{\square\prime} \right\rangle . \tag{S3.8}$$

S3.1 The thermodynamic double virial estimator

This estimator is obtained by deriving the centroid-virial thermodynamic estimator of the energy:

$$\frac{C_V^{\mathrm{TD}}}{k_B \beta^2} = \left\langle \mathcal{E}^{\mathrm{TD}} \cdot \mathcal{E}^{\mathrm{TD}} \right\rangle - \left\langle \mathcal{E}^{\mathrm{TD}} \right\rangle \cdot \left\langle \mathcal{E}^{\mathrm{TD}} \right\rangle + \left\langle \mathcal{E}^{\mathrm{TD}\prime} \right\rangle . \tag{S3.9}$$

Since $\mathcal{E}^{\mathrm{TD}}$ contains second order derivatives of the potential, $\mathcal{E}^{\mathrm{TD}\prime}$ explicitly depends on the third order derivatives of the potential, making its direct evaluation prohibitive for most interatomic potentials. A more practical way of evaluating it – as prescribed by Yamamoto[13] – is the following temperature derivative:

$$\mathcal{E}^{\mathrm{TD}\prime} = \frac{3N}{2\beta^2} - \frac{1}{P} \sum_{j=1}^{P} \frac{\partial^2}{\partial \beta'^2} [\beta' \, V^{\mathrm{sc}} \left(\bar{\mathbf{q}} + \frac{\beta'}{\beta} \left(\mathbf{q}^{(j)} - \bar{\mathbf{q}} \right) \right)] \Bigg|_{\beta'=\beta} , \tag{S3.10}$$

where the second order derivative is estimated through a centred finite difference: $f''(x) = \lim_{\epsilon \to 0} \frac{f(x+\epsilon)+f(x-\epsilon)-2f(x)}{\epsilon^2}$. This allows one to compute the double virial thermodynamic estimator of the heat capacity from $2P$ additional evaluations of the energy and forces.

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S3.2 The operator double virial estimator

Another way of calculating a double virial estimator of the heat capacity – which we derive here – is by deriving the operator estimator of the energy:

$$\frac{C_V^{\mathrm{OP}}}{k_B\beta^2} = \left\langle \mathcal{E}^{\mathrm{OP}} \cdot \mathcal{E}^{\mathrm{TD}} \right\rangle - \left\langle \mathcal{E}^{\mathrm{OP}} \right\rangle \cdot \left\langle \mathcal{E}^{\mathrm{TD}} \right\rangle + \left\langle \mathcal{E}^{\mathrm{OP}\prime} \right\rangle, \tag{S3.11}$$

where

$$\left\langle E^{\mathrm{OP}\prime} \right\rangle = \frac{3N}{2\beta^2} + \frac{1}{\beta P}\sum_{j=1}^{P/2} (\mathbf{q}^{(2j-1)} - \bar{\mathbf{q}}) \cdot \mathbf{f}_{dcv}^{(2j-1)} \tag{S3.12}$$

$$\mathbf{f}_{dcv}^{(2j-1)} = \frac{3}{2}\mathbf{f}^{(2j-1)} + \frac{1}{2}\frac{\partial \mathbf{f}^{(2j-1)}}{\partial \mathbf{q}^{(2j-1)}} \cdot (\mathbf{q}^{(2j-1)} - \bar{\mathbf{q}}). \tag{S3.13}$$

Here, the double virial force $\mathbf{f}_{dcv}$ depends on the second order derivative of the potential. It can however be computed through the following finite difference:

$$\mathbf{f}_{dcv}^{(2j-1)} = \frac{3}{2}\mathbf{f}^{(2j-1)} + \frac{1}{2}\lim_{\epsilon\to 0}\epsilon^{-1}\left(\mathbf{f}^{(2j-1)}\Big|_{\mathbf{q}^{(2j-1)}+\epsilon(\mathbf{q}^{(2j-1)}-\bar{\mathbf{q}})} - \mathbf{f}^{(2j-1)} \right). \tag{S3.14}$$

Note that the double virial force only exists for the odd replicas and is therefore computed with just $P/2$ additional force evaluations, instead of the $2P$ evaluations required by the estimator of Yamamoto.

S4 Force field derivation and validation

The first-principles force fields used to model our systems are derived using `QuickFF`,[14,15] a software package developed by some of the present authors. Within this protocol, the quantum mechanical potential energy surface (PES) is approximated by a sum of analytical functions of the nuclear coordinates that describe the covalent (cov) and noncovalent

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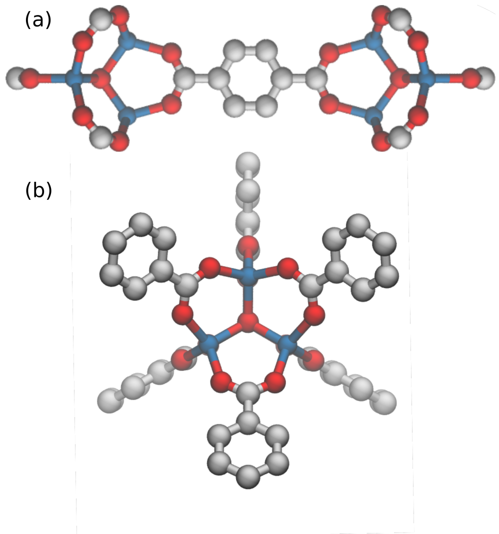

Figure S1: Molecular representation of the cluster models that were used for the organic linker (top) and inorganic brick (bottom) to generate the required first-principles input data.

(noncov) interactions. The latter is composed of electrostatic and van der Waals interactions.

$$V^{FF} = \underbrace{V_{bond} + V_{bend} + V_{oopd} + V_{torsion} + V_{cross}}_{V_{cov}} + V_{ei} + V_{vdW}. \tag{S4.1}$$

The force field used in this work and a guest-loaded structure of methane are added as additional files, which enables one to perform similar simulations with `Yaff`.[16]

S4.1 First-principles cluster data

Just as in our previous work on MOFs,[14,17] we use isolated cluster models to generate the required first-principles input data. For MOF-5, we construct a cluster for the inorganic brick and a cluster for the organic linker, as originally proposed by Schmid and co-workers.[18] The DFT optimized structures (of the MOF) are shown in Figure S1.

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S4.2 Covalent interactions

The covalent interactions – which mimic the chemical bonds between the atoms – are approximated by different terms as a function of the internal coordinates (bonds, bends, out-of-plane distances, and dihedrals). The unknown force field parameters are fitted following the `QuickFF` procedure.[14] We use an extended version[15] to include anharmonic bond and bend terms,[19] which are important when modeling nuclear quantum effects and the thermal expansion of MOFs.[15,20] The anharmonic bond and bend terms are given by:

$$V_{\text{bond}}^{ij} = \frac{K_{ij}}{2}\left(r_{ij} - r_{0,ij}\right)^2 \left[1 - \alpha\left(r_{ij} - r_{0,ij}\right) + \frac{7}{12}\cdot\alpha^2\cdot\left(r_{ij} - r_{0,ij}\right)^2\right], \tag{S4.2}$$

$$\begin{aligned} V_{\text{bend}}^{ijk} = \frac{K_{ijk}}{2}\left(\theta_{ijk} - \theta_{0,ijk}\right)^2 \Big[1 &- a_1\left(\theta_{ijk} - \theta_{0,ijk}\right) + a_2\cdot\left(\theta_{ijk} - \theta_{0,ijk}\right)^2 \\ &- a_3\cdot\left(\theta_{ijk} - \theta_{0,ijk}\right)^3 + a_4\left(\theta_{ijk} - \theta_{0,ijk}\right)^4\Big], \end{aligned} \tag{S4.3}$$

with $\alpha = 2.55$ Å^{-1}, $a_1 = 0.014$ deg^{-1}, $a_2 = 5.6\cdot10^{-5}$ deg^{-2}, $a_3 = 7\cdot10^{-7}$ deg^{-3}, and $a_4 = 2.2\cdot10^{-8}$ deg^{-4}. For methane, only harmonic bonds and bends were used. We also add cross terms between the bonds and bends to improve the correspondence with the first-principles data.[15] The included cross terms are:

- angle stretch-stretch terms (ASS) between neigboring bonds, *i.e.* part of the same angle
- angle stretch-angle terms (ASA).

The mathematical expression of these terms is given by

$$V_{\text{ASS}}^{ijk} = K_{ijk}^{\text{ASS}}\left(r_{ij} - r_{0,ij}\right)\left(r_{jk} - r_{0,jk}\right) \tag{S4.4}$$

$$V_{\text{ASA}}^{ijk} = \left[K_{ijk}^{\text{ASA1}}\left(r_{ij} - r_{0,ij}\right) + K_{ijk}^{\text{ASA2}}\left(r_{jk} - r_{0,jk}\right)\right]\left(\theta_{ijk} - \theta_{0,ijk}\right). \tag{S4.5}$$

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The out-of-plane distances are described using a harmonic potential:

$$V_{oopd}^{ijkl} = \frac{K_{ijkl}}{2} \left(d_{ijkl} - d_{0,ijkl}\right)^2. \tag{S4.6}$$

This is a four-atom interaction, in which the internal coordinate is the distance between the central atom and the plane determined by its three neighbors. The fourth covalent term is the dihedral energy term. Here, a cosine term is used as a function of the dihedral angle, including the multiplicity m_ϕ of the dihedral angle:

$$V_{torsion}^{ijkl} = \frac{K_{ijkl}}{2} \left[1 - \cos\big(m_\phi(\phi_{ijkl} - \phi_{0,ijkl})\big)\right]. \tag{S4.7}$$

The unknown parameters in all terms (force constants, rest values, and multiplicities) in the covalent energy expression can be estimated directly with `QuickFF`.

S4.3 Electrostatic interactions

The electrostatic interactions are modeled by a Coulomb interaction between Gaussian charge distributions,[21] which allow to include all pairwise interactions. The atomic charges q_i are derived with the Minimal Basis Iterative Stockholder (MBIS) partitioning scheme[22] as implemented in `Horton`.[23]

$$V_{ei} = \frac{1}{2} \sum_{\substack{i,j=1 \\ (i \neq j)}} \frac{q_i q_j}{4\pi\epsilon_0 r_{ij}} \operatorname{erf}\left(\frac{r_{ij}}{d_{ij}}\right) \tag{S4.8}$$

Gaussian charge distributions are used with a total charge q_i and radius d_i, centered on atom i. The mixed radius of the Gaussian charges,[21] d_{ij}, is given by $\sqrt{d_i^2 + d_j^2}$. The interaction depends on the distance r_{ij} between the two atoms.

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S4.4 Van der Waals interactions

The van der Waals interactions are described by the MM3-Buckingham model[24,25] up to a finite cutoff and are supplemented with tail corrections.[26] This choice of van der Waals interactions between the framework and the guests produces reasonably accurate results for the adsorption energies and adsorption isotherms.[27,28] The van der Waals interactions are only included in the force field of the periodic structure (*i.e.* they are not included in the fitting procedure of the covalent force field parameters).

$$V_{vdW} = \epsilon_{ij} \left[1.84 \cdot 10^5 \exp\left(-12 \frac{r}{\sigma_{ij}}\right) - 2.25 \left(\frac{\sigma_{ij}}{r}\right)^6 \right] \tag{S4.9}$$

The two parameters σ_{ij} and ϵ_{ij} are the equilibrium distance and the well depth of the potential. These parameters are typically determined with empirical mixing rules for the interaction between atom i and atom j:

$$\sigma_{ij} = \sigma_i + \sigma_j \qquad \text{and} \qquad \epsilon_{ij} = \sqrt{\epsilon_i \epsilon_j} \tag{S4.10}$$

and these parameters were taken from the MM3 force field for every atom and are tabulated in Ref. 25. In MM3, the 1-2 and 1-3 interactions are discarded to avoid a strong overestimation of the repulsion terms.

S4.5 Validation

Previously, similar `QuickFF` force fields have been successfully applied to a wide range of MOFs for various properties such as lattice parameters, bulk moduli, elastic constants, and free energy profiles.[17,29–31] To validate the force fields in this work, we compared the first-principles cluster results with those obtained by the new force fields. The force fields succeed in reproducing various internal coordinates of the cluster models, together with the

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first-principles frequencies. The geometry optimized structure at 0 K with this force field transferred to the periodic structure has a lattice constant of 26.4 Å, which is in good agreement with earlier studies (*e.g.* Refs. 14,18,32).

S5 Methane adsorption in MOF-5

In this work, we investigate the archetypal case of the well-known MOF-5[32,33] in the presence and absence of adsorbed methane. Methane adsorption in this framework has been the topic of detailed experimental and theoretical works (*e.g.* Refs. 34–36), and adsorption isotherms at various temperatures up to high gas pressures are available (*e.g.* Refs. 37,38).

We performed Grand Canonical Monte Carlo (GCMC) simulations with `RASPA`[39] using our newly derived force field. These simulations were performed with a fixed framework and rigid methane molecules, and the guest-host and guest-guest interactions were modeled through the noncovalent interactions. In Figure S2, the simulated isotherm at room temperature is compared with experimental data at room temperature. Our force field underestimates the high-pressure saturation loading at various temperatures (Figure S3), which is a property that is extremely sensitive to the underlying potential energy surface.[28]

To illustrate this, we performed additional simulations using a different force field for methane (Figure S3). We used the TraPPE force field to describe the guest-guest interactions, and the guest-host interactions are modeled via the TraPPE parameters for methane[40] and UFF parameters[41] for MOF-5. In case of TraPPE, methane is modeled as a single site. This single site is not charged, so the only guest-host and guest-guest interactions that are present are from the van der Waals type. We find when compared to experiment that UFF-TraPPE overestimates the loading and our MM3-MBIS force field underestimates the loading at every temperature.

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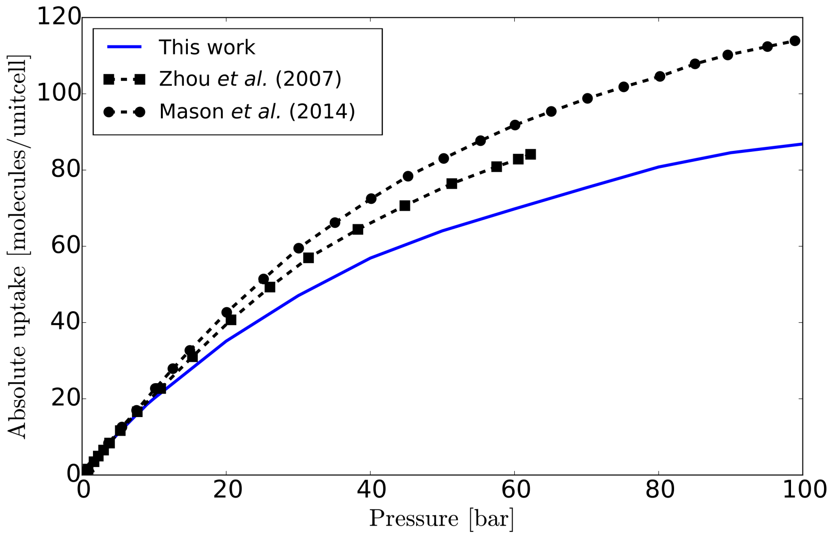

Figure S2: A comparison between the experimental and our simulated MM3-MBIS adsorption isotherms at room temperature. The experimental data[37,38] were converted to methane molecules per unit cell.

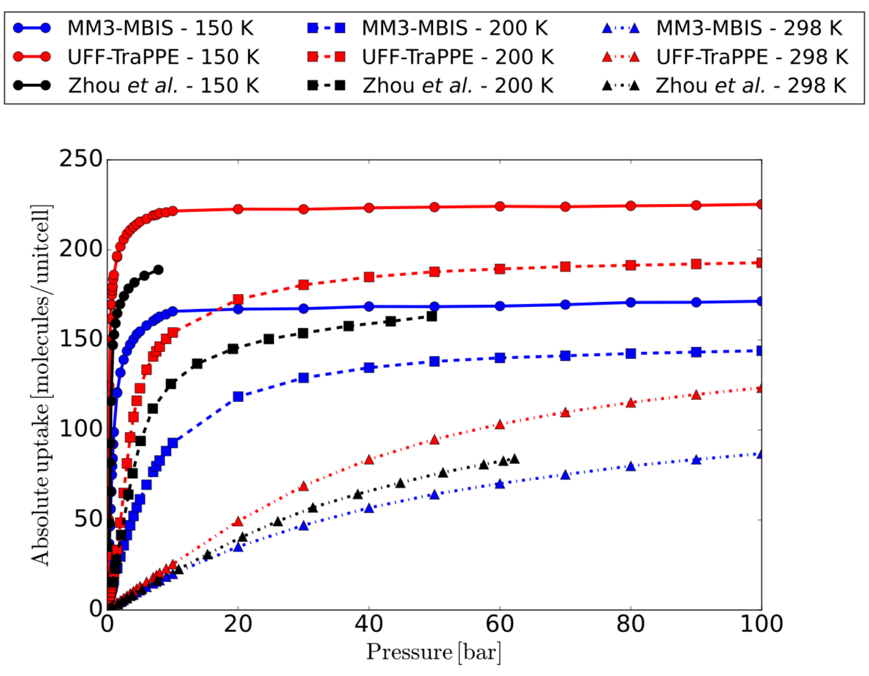

Figure S3: A comparison between the experimental and our simulated adsorption isotherms at different temperatures. The experimental data[37] were converted to methane molecules per unit cell.

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S6 Heat capacity

S6.1 Harmonic approximation

We computed the Hessian, *i.e.* the second order derivatives of the energy with respect to the atomic coordinates, with `Yaff`[16] and performed a normal mode analysis with `TAMkin`.[42] Using the normal mode frequencies ω, we evaluate the isochoric heat capacity using the harmonic approximation:

$$C_V(T) = k_B \sum_{\omega} \left(\frac{\hbar\omega}{k_B T}\right)^2 \frac{e^{\frac{\hbar\omega}{k_B T}}}{\left(e^{\frac{\hbar\omega}{k_B T}} - 1\right)^2}. \tag{S6.1}$$

In Figure S4, we show the results for empty MOF-5 obtained with our first-principles-derived force field and with the generic UFF4MOF force field.[43] This indicates that even with a generic potential energy surface model results of the same order of magnitude can be obtained for the heat capacity of the empty framework.

We also computed the harmonic approximation for MOF-5 loaded with 100 methane molecules per conventional unit cell. We performed optimizations starting from 5 different Monte Carlo snapshots. Not surprisingly, we do not find exactly the same structures after optimizations due to the presence of multiple minima on this complex potential energy surface. However, the influence of these different configurations on the harmonic approximation of the heat capacity is very limited, as shown in Figure S5.

S6.2 Comparison of barostats and software packages

Figure S6 shows the comparison of the temperature dependent classical isobaric heat capacity computed using the isotropic and flexible Martyna-Tobias-Klein (MTK) barostat as implemented in `Yaff`[44–46] and the isotropic Bussi-Zykova-Parrinello (BZP) barostat as im-

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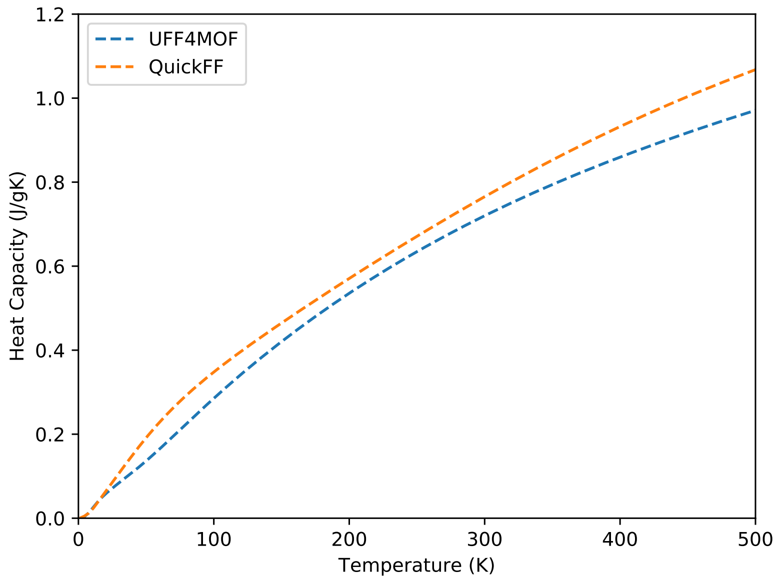

Figure S4: The heat capacity of the empty MOF-5 framework obtained using the harmonic approximation. We compare the results of our first-principles derived force field with UFF4MOF.[43]

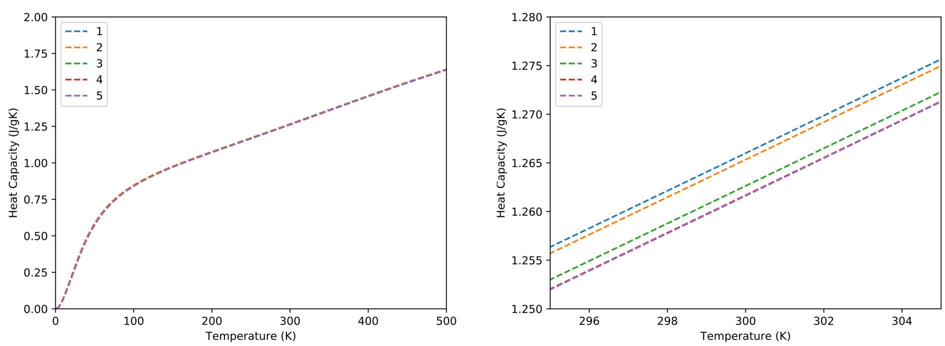

Figure S5: The heat capacity obtained using the harmonic approximation for MOF-5 with 100 methane molecules per unit cell. The influence of different configurations of methane in MOF-5 is limited.

plemented in `i-PI`[8,9] for a loading of 100 methane molecules in MOF-5. In other words, we compare the applicability and the implementation of the $N\mathcal{P}(\boldsymbol{\sigma}_a = \mathbf{0})T$ and $N\mathcal{P}(\boldsymbol{h}_0)T$ ensembles in `Yaff` and `i-PI`. We find that the isobaric heat capacity for this rigid MOF is not significantly affected by cell shape sampling. We have therefore only implemented an isotropic barostat for the Suzuki-Chin Hamiltonian as it suffices for the current investigation.

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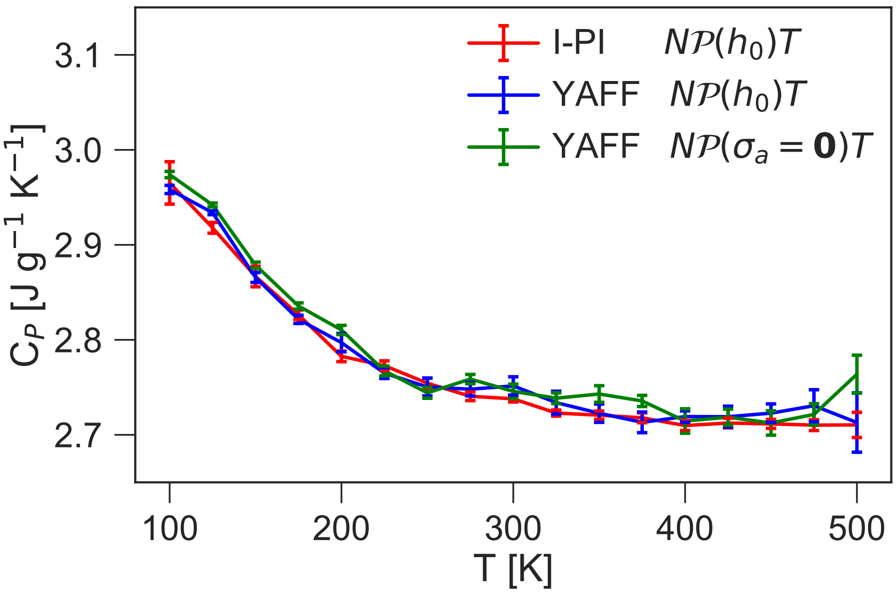

Figure S6: The temperature dependence of the classical isobaric heat capacity obtained in the $N\mathscr{P}(\boldsymbol{h}_0)T$ ensemble with `i-PI` (red) and `Yaff` (blue) and in the $N\mathscr{P}(\boldsymbol{\sigma}_a = \mathbf{0})T$ ensemble with `Yaff` (green) for MOF-5 with 100 methane molecules per unit cell.

S6.3 Force-field energy decomposition

To gain insight into the factors determining the heat capacity of guest-loaded MOFs, we decomposed the force-field energy to analyze the different contributions to the heat capacity. We investigated which covalent and noncovalent interactions store thermal energy and thus determine the heat capacity. This decomposition is made possible by the use of a force field approach. Thereto, we revisited our MD trajectories. The classical total energy of the system at each time step is given by the sum of the kinetic energy T and the potential energy V. We do not decompose the classical kinetic energy, which is given by $\frac{3}{2}Nk_BT$. The potential energy in the classical MD trajectory can easily be decomposed in terms of covalent or noncovalent contributions, or in host, host-guest, and guest-guest interactions. Practically, we evaluated the potential energy along the MD trajectory of (1) the framework atoms in the absence of the guests and (2) the methane molecules in the absence of the framework. The difference between the total potential energy and the host and guest-guest

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interactions then gives the host-guest contribution. The kinetic contribution is splitted by adding $\frac{3}{2}N_{host}k_BT$ and $\frac{3}{2}N_{guest}k_BT$ to the host and guest-guest interactions, respectively. By applying a finite-differences strategy on these host, host-guest and, guest-guest energies at different temperatures, we are thus able to determine the contribution to the total heat capacity of the host, host-guest, and guest-guest interactions. Similarly, we can split the potential energy in the different force-field energy terms by printing out the separate contributions every time step. For this decomposition, we made the extra approximation of adding the kinetic contribution completely to the covalent terms.

The above discussion holds for the classical MD results. For PIMD, however, we need to slightly adapt our approach. We focus on the quantum mechanical virial energy estimator in PIMD given by

$$E = \frac{3}{2}Nk_BT + \frac{1}{2P}\sum_{i=1}^{N}\sum_{k=1}^{P}\left(\mathbf{r}_i^{(k)} - \mathbf{r}_i^{(c)}\right)\cdot\frac{\partial V}{\partial \mathbf{r}_i^{(k)}} + \frac{1}{P}\sum_{k=1}^{P}V. \tag{S6.2}$$

The first two terms determine the quantum kinetic energy, while the last term represents the potential energy. Due to the quantum virial theorem, it is possible to write the quantum kinetic energy in terms of the derivative of the potential energy, *i.e.* the forces, multiplied with the distance between the bead and the centroid.

This expression can be evaluated by taking snapshots from the quantum ensemble that we sampled with our PIMD simulations, which is done by taking the odd beads of our Suzuki-Chin simulations. Analogously to the classical simulations, we evaluate the forces for the host and guest-guest interactions by removing the guest and host atoms, respectively, and determine the potential energy in a similar way. We obtain contributions to the heat capacity after taking the numerical derivative of these energies. We again make the extra approximation of adding the constant kinetic contribution completely to the covalent terms.

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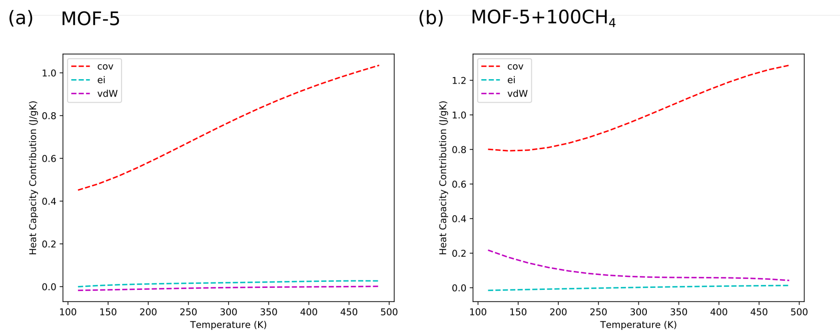

Figure S7: Force-field decomposition of the contributions to the heat capacity for: (a) MOF-5 (b) MOF-5 + 100 CH_4.

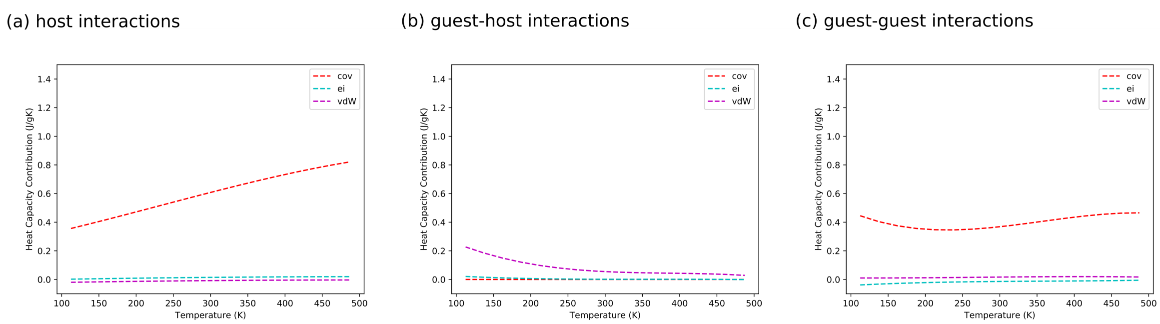

Figure S8: Force-field decomposition of the contributions to the heat capacity for MOF-5 + 100 CH_4.

S6.4 Volumetric heat capacity

Figure S10 shows the temperature dependence of the heat capacity per unit volume or the volumetric heat capacity which is computed as the ratio of the isobaric heat capacity $C_P(T)$ and the molar volume $\mathcal{V}(T)$. A non-monotonic behavior with a minimum near 175 K is observed.

S6.5 Comparison with UFF/TraPPE for guest-guest and guest-host interactions

In addition, we also used the noncovalent UFF-TraPPE force field (see Section S5) to perform PIMD simulations. The covalent and electrostatic interactions of MOF-5 are modeled with the same parameters as described above, but the noncovalent van der Waals model is replaced

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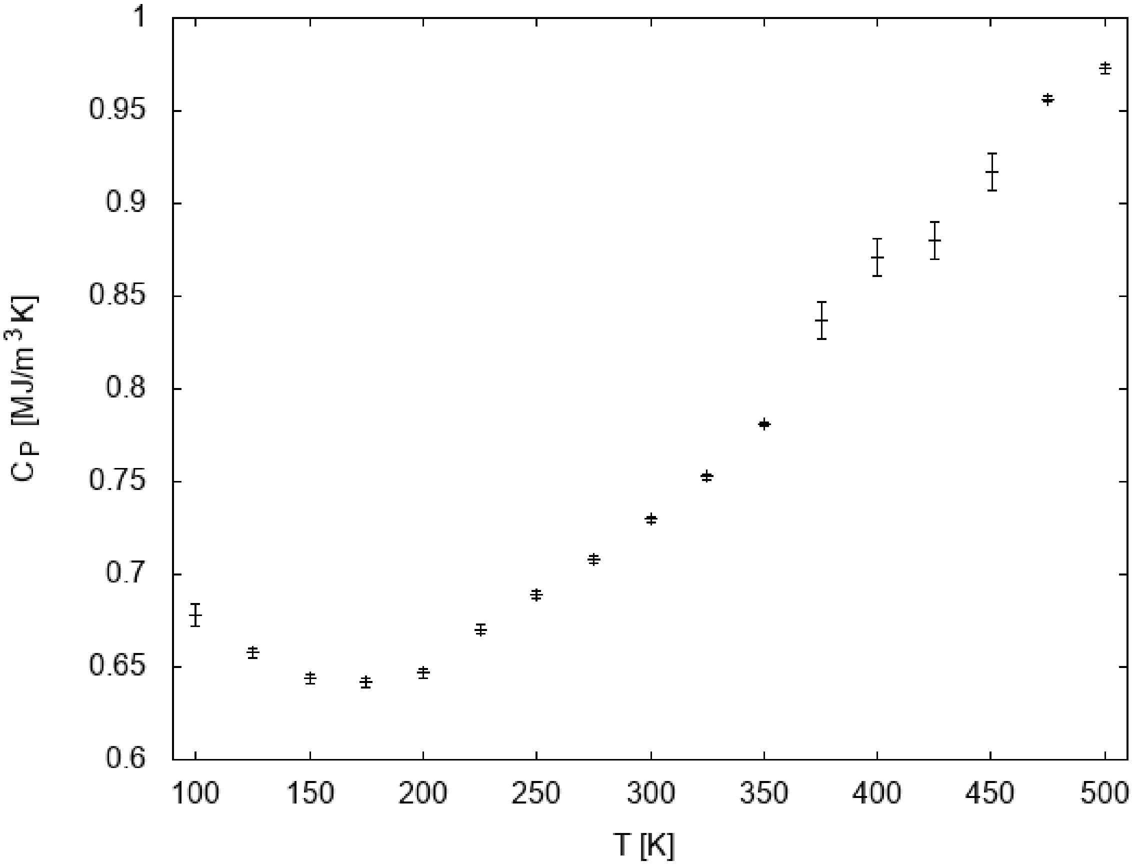

Figure S9: The temperature dependence of the volumetric heat capacity of MOF-5 + 100 CH_4.

with UFF-TraPPE. Our results display the same qualitative behavior as our simulations using the QuickFF/MM3-MBIS force field used in the main article.

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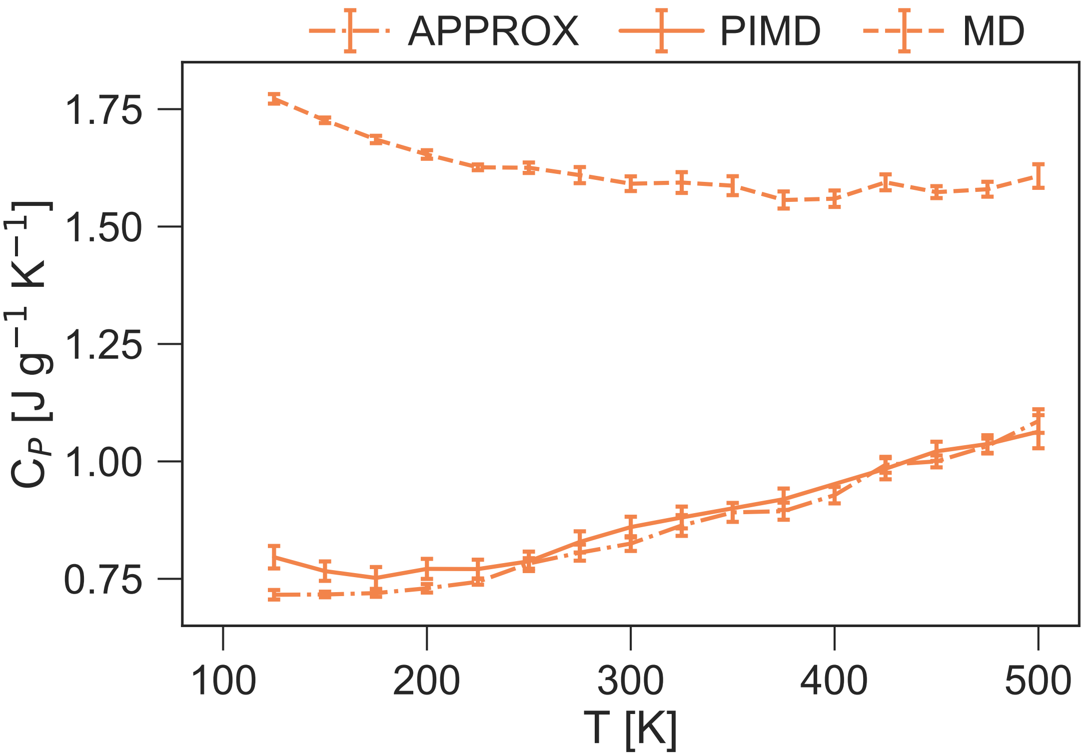

Figure S10: Temperature dependence of the isobaric heat capacity of MOF-5 with 100 molecules of methane computed using classical MD (dashed), PIMD (dotted) and the approximation proposed in the work (dot dashed). The noncovalent van der Waals model is replaced with UFF-TraPPE. .

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